

Cloud Computing Practical Examination

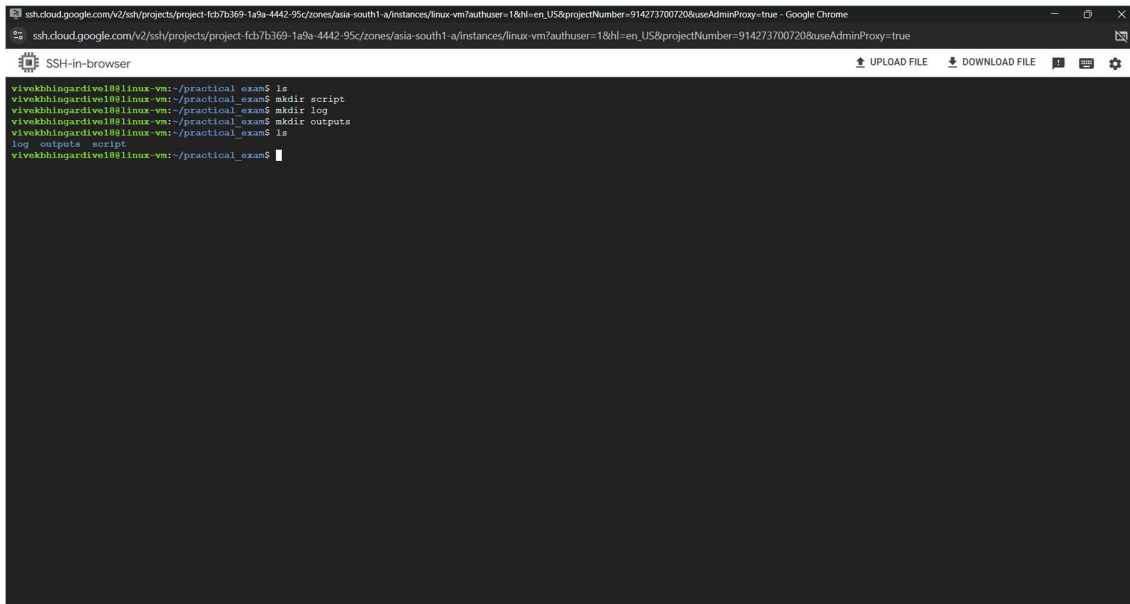
Part A – Cloud & Linux VM Setup

Task 1: Create a Linux Virtual Machine with Public IP and SSH access enabled.

Task 2: Connect to the VM using SSH and capture screenshots.

Task 3: Create practical_exam/scripts, practical_exam/outputs and practical_exam/logs.

Soln.:

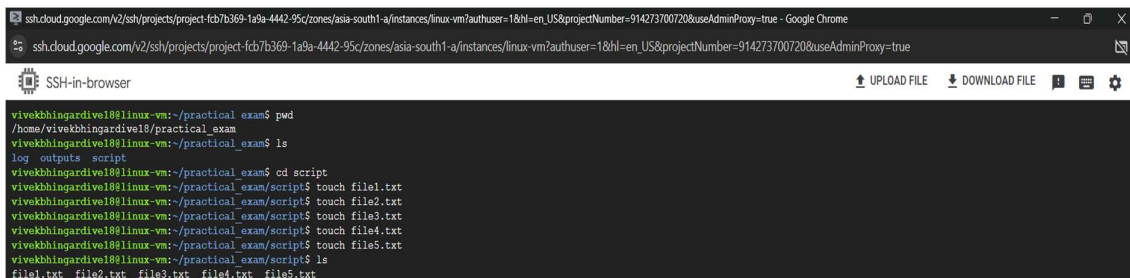


```
ssh.cloud.google.com/v2/ssh/projects/project-fcb7b369-1a9a-4442-95c/zones/asia-south1-a/instances/linux-vm?authuser=1&hl=en_US&projectNumber=914273700720&useAdminProxy=true - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/project-fcb7b369-1a9a-4442-95c/zones/asia-south1-a/instances/linux-vm?authuser=1&hl=en_US&projectNumber=914273700720&useAdminProxy=true
SSH-in-browser
vivekhhingardive18@linux-vm:~/practical_exam$ ls
vivekhhingardive18@linux-vm:~/practical_exam$ mkdir script
vivekhhingardive18@linux-vm:~/practical_exam$ mkdir log
vivekhhingardive18@linux-vm:~/practical_exam$ mkdir outputs
vivekhhingardive18@linux-vm:~/practical_exam$ ls
log  outputs  script
vivekhhingardive18@linux-vm:~/practical_exam$
```

Part B – Linux Commands

Task 4: Display current working directory, current user and hostname. Save output.

Task 5: Create 5 files inside scripts folder.



```
ssh.cloud.google.com/v2/ssh/projects/project-fcb7b369-1a9a-4442-95c/zones/asia-south1-a/instances/linux-vm?authuser=1&hl=en_US&projectNumber=914273700720&useAdminProxy=true - Google Chrome
ssh.cloud.google.com/v2/ssh/projects/project-fcb7b369-1a9a-4442-95c/zones/asia-south1-a/instances/linux-vm?authuser=1&hl=en_US&projectNumber=914273700720&useAdminProxy=true
SSH-in-browser
vivekhhingardive18@linux-vm:~/practical_exam$ pwd
/home/vivekhhingardive18/practical_exam
vivekhhingardive18@linux-vm:~/practical_exam$ ls
log  outputs  script
vivekhhingardive18@linux-vm:~/practical_exam$ cd script
vivekhhingardive18@linux-vm:~/practical_exam/script$ touch file1.txt
vivekhhingardive18@linux-vm:~/practical_exam/script$ touch file2.txt
vivekhhingardive18@linux-vm:~/practical_exam/script$ touch file3.txt
vivekhhingardive18@linux-vm:~/practical_exam/script$ touch file4.txt
vivekhhingardive18@linux-vm:~/practical_exam/script$ touch file5.txt
vivekhhingardive18@linux-vm:~/practical_exam/script$ ls
file1.txt  file2.txt  file3.txt  file4.txt  file5.txt
```

```
ssh.cloud.google.com/v2/ssh/projects/project-fcb7b369-1a9a-4442-95c/zones/asia-south1-a/instances/linux-vm...
SSH-in-browser
vivekbhingardive18@linux-vm:~/practical_exam$ cd script
vivekbhingardive18@linux-vm:~/practical_exam/script$ mkdir file1.sh
vivekbhingardive18@linux-vm:~/practical_exam/script$ mkdir file2.sh
vivekbhingardive18@linux-vm:~/practical_exam/script$ mkdir file3.sh
vivekbhingardive18@linux-vm:~/practical_exam/script$ mkdir file3.sh
mkdir: cannot create directory 'file3.sh': File exists
vivekbhingardive18@linux-vm:~/practical_exam/script$ mkdir file4.sh
vivekbhingardive18@linux-vm:~/practical_exam/script$ mkdir file5.sh
vivekbhingardive18@linux-vm:~/practical_exam/script$ ls
file1.sh  file1_outputs.txt  file2.txt  file3.sh  file4.sh  file5.sh  outputs
file1.txt  file2.sh  file2_outputs.txt  file3.txt  file4.txt  file5.txt
vivekbhingardive18@linux-vm:~/practical_exam/script$
```

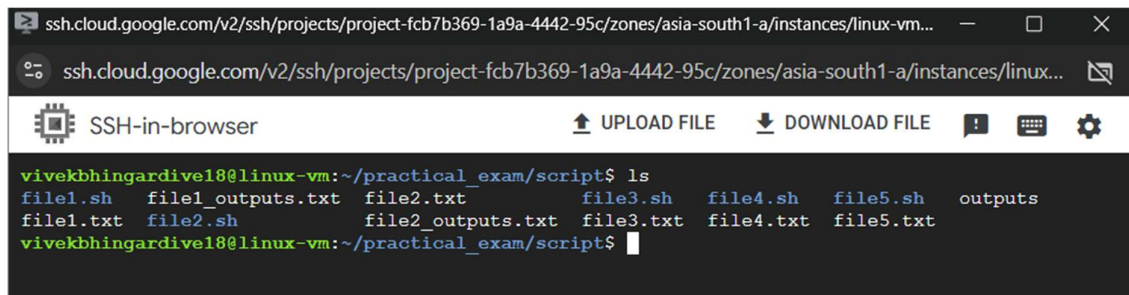
Task 6: Copy two files into outputs folder.

Soln. :

```
ssh.cloud.google.com/v2/ssh/projects/project-fcb7b369-1a9a-4442-95c/zones/asia-south1-a/instances/linux-vm...
SSH-in-browser
vivekbhingardive18@linux-vm:~/practical_exam$ cd script
vivekbhingardive18@linux-vm:~/practical_exam/script$ mkdir file1.sh
vivekbhingardive18@linux-vm:~/practical_exam/script$ mkdir file2.sh
vivekbhingardive18@linux-vm:~/practical_exam/script$ mkdir file3.sh
vivekbhingardive18@linux-vm:~/practical_exam/script$ mkdir file3.sh
mkdir: cannot create directory 'file3.sh': File exists
vivekbhingardive18@linux-vm:~/practical_exam/script$ mkdir file4.sh
vivekbhingardive18@linux-vm:~/practical_exam/script$ mkdir file5.sh
vivekbhingardive18@linux-vm:~/practical_exam/script$ ls
file1.sh  file1_outputs.txt  file2.txt  file3.sh  file4.sh  file5.sh  outputs
file1.txt  file2.sh  file2_outputs.txt  file3.txt  file4.txt  file5.txt
vivekbhingardive18@linux-vm:~/practical_exam/script$ cp outputs/file1.txt /
cp: cannot stat 'outputs/file1.txt': Not a directory
vivekbhingardive18@linux-vm:~/practical_exam/script$ cd ..
vivekbhingardive18@linux-vm:~/practical_exam$ cp scripts/file1.txt outputs
cp: cannot stat 'scripts/file1.txt': No such file or directory
vivekbhingardive18@linux-vm:~/practical_exam$ cp script/file1.txt outputs
vivekbhingardive18@linux-vm:~/practical_exam$ cp script/file2.txt outputs
vivekbhingardive18@linux-vm:~/practical_exam$ cd outputs
vivekbhingardive18@linux-vm:~/practical_exam/outputs$ ls
file1.txt  file2.txt
vivekbhingardive18@linux-vm:~/practical_exam/outputs$
```

Task 7: Rename one file.

Soln :

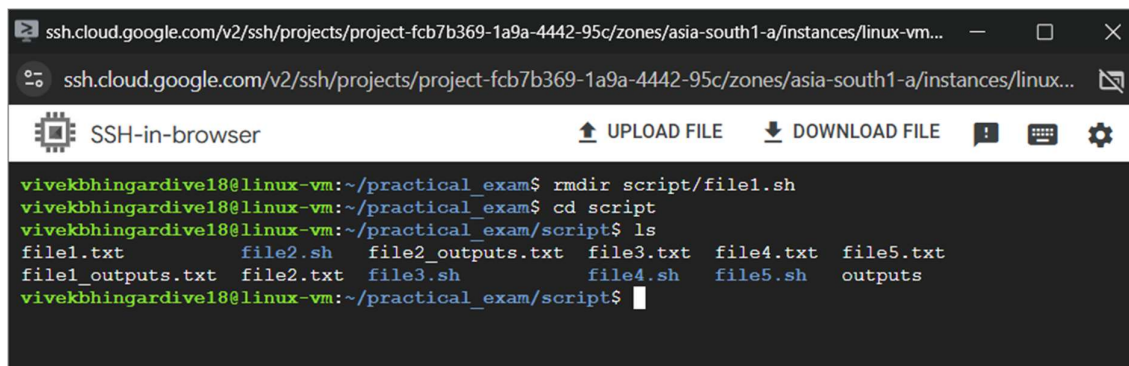


```
ssh.cloud.google.com/v2/ssh/projects/project-fcb7b369-1a9a-4442-95c/zones/asia-south1-a/instances/linux-vm...  
SSH-in-browser  
vivekbhingardive18@linux-vm:~/practical_exam/script$ ls  
file1.sh  file1_outputs.txt  file2.txt      file3.sh  file4.sh  file5.sh  outputs  
file1.txt  file2.sh          file2_outputs.txt  file3.txt  file4.txt  file5.txt  
vivekbhingardive18@linux-vm:~/practical_exam/script$
```

Explanation : I renamed file1.txt to file1_output.txt

Task 8: Delete one file.

Soln. :



```
ssh.cloud.google.com/v2/ssh/projects/project-fcb7b369-1a9a-4442-95c/zones/asia-south1-a/instances/linux-vm...  
SSH-in-browser  
vivekbhingardive18@linux-vm:~/practical_exam$ rmdir script/file1.sh  
vivekbhingardive18@linux-vm:~/practical_exam$ cd script  
vivekbhingardive18@linux-vm:~/practical_exam/script$ ls  
file1.txt      file2.sh  file2_outputs.txt  file3.txt  file4.txt  file5.txt  
file1_outputs.txt  file2.txt  file3.sh          file4.sh  file5.sh  outputs  
vivekbhingardive18@linux-vm:~/practical_exam/script$
```

Task 9: Display hidden files.

soln. :

Task 10: Display disk usage information.

Task 11: Display memory usage information.

Task 12: Display running processes.

Task 13: Find all .sh files inside practical_exam.

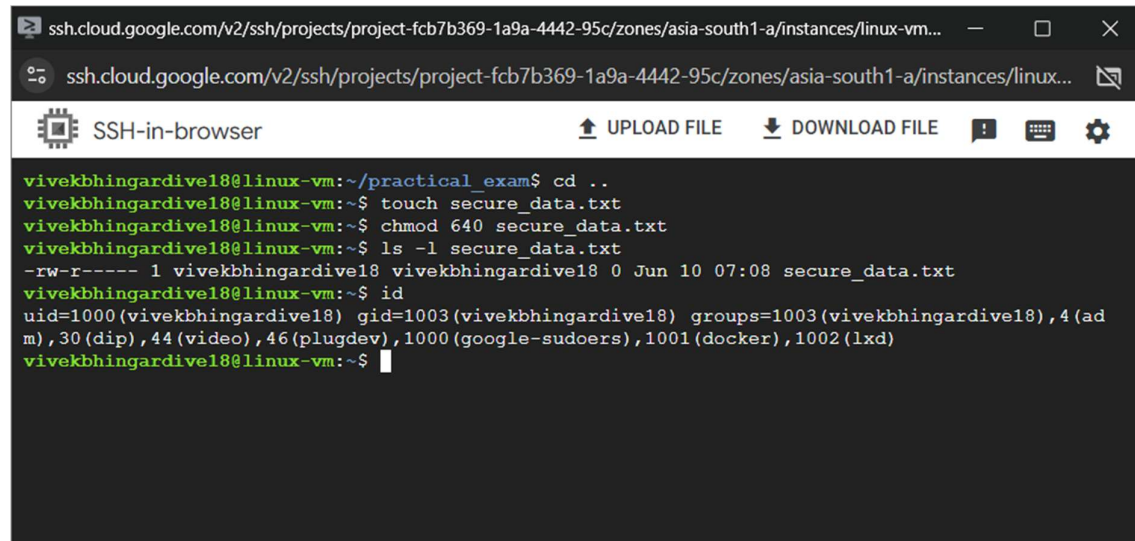
Part C – Permissions & Users

Task 14: Create secure_data.txt and apply permission 640.

Task 15: Demonstrate Owner, Group and Others using commands.

Task 16: Display User ID, Group ID and assigned groups.

Soln. :

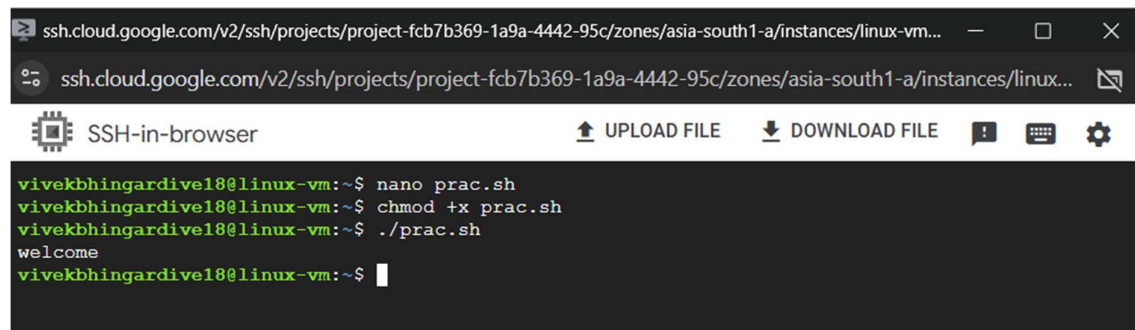


```
ssh.cloud.google.com/v2/ssh/projects/project-fcb7b369-1a9a-4442-95c/zones/asia-south1-a/instances/linux-vm...  
SSH-in-browser  
vivekbhingardive18@linux-vm:~/practical_exam$ cd ..  
vivekbhingardive18@linux-vm:~$ touch secure_data.txt  
vivekbhingardive18@linux-vm:~$ chmod 640 secure_data.txt  
vivekbhingardive18@linux-vm:~$ ls -l secure_data.txt  
-rw-r----- 1 vivekbhingardive18 vivekbhingardive18 0 Jun 10 07:08 secure_data.txt  
vivekbhingardive18@linux-vm:~$ id  
uid=1000(vivekbhingardive18) gid=1003(vivekbhingardive18) groups=1003(vivekbhingardive18),4(adm),30(dip),44(video),46(plugdev),1000(google-sudoers),1001(docker),1002(lxd)  
vivekbhingardive18@linux-vm:~$
```

Part D – Shell Scripting

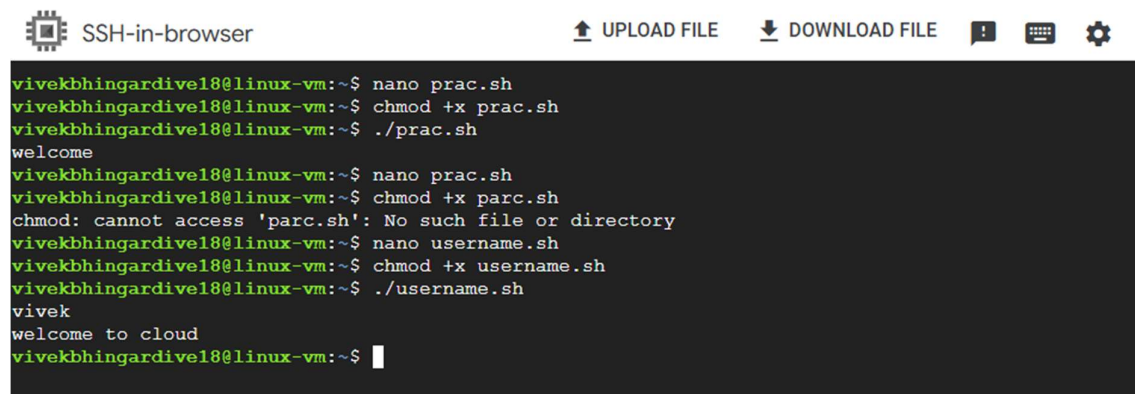
Task 17: Script to print Welcome to Cloud Computing.

Soln. :



```
ssh.cloud.google.com/v2/ssh/projects/project-fcb7b369-1a9a-4442-95c/zones/asia-south1-a/instances/linux-vm...  
SSH-in-browser  
vivekbhingardive18@linux-vm:~$ nano prac.sh  
vivekbhingardive18@linux-vm:~$ chmod +x prac.sh  
vivekbhingardive18@linux-vm:~$ ./prac.sh  
welcome  
vivekbhingardive18@linux-vm:~$
```

Task 18: Script accepting username and displaying Welcome
soln. :

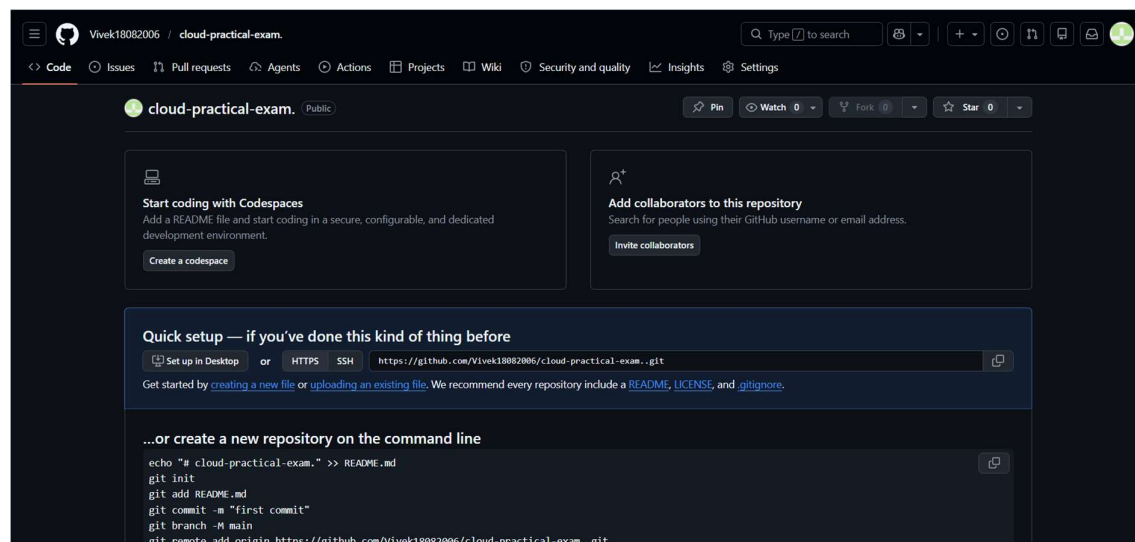


The image shows a terminal window titled "SSH-in-browser" with a dark background and green text. The user is logged in as vivekbhingardive18@linux-vm. The terminal shows the following commands and output:

```
vivekbhingardive18@linux-vm:~$ nano prac.sh
vivekbhingardive18@linux-vm:~$ chmod +x prac.sh
vivekbhingardive18@linux-vm:~$ ./prac.sh
welcome
vivekbhingardive18@linux-vm:~$ nano prac.sh
vivekbhingardive18@linux-vm:~$ chmod +x parc.sh
chmod: cannot access 'parc.sh': No such file or directory
vivekbhingardive18@linux-vm:~$ nano username.sh
vivekbhingardive18@linux-vm:~$ chmod +x username.sh
vivekbhingardive18@linux-vm:~$ ./username.sh
vivek
welcome to cloud
vivekbhingardive18@linux-vm:~$
```

Part F – Git & Submission

Task 31: Create GitHub repository cloud-practical-exam.
Soln. :



Task 32: Initialize Git in practical_exam directory.

Task 33: Add all files.

Task 34: Commit changes with message Cloud Practical Exam Submission.

Task 35: Push files to GitHub.

Task 36: Verify repository contains scripts, outputs, screenshots and documentation.

Soln. :

Vivek18082006 / cloud-practical-exam.

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Vivek18082006

Vivek Bhingardive